

MSDS 6371 – Statistical Foundations for Data Science
Individual Project Report

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Abstract

This project analyzes housing price dynamics in the Ames, Iowa dataset through two complementary statistical investigations. Analysis 1 examines whether the relationship between above-ground living area and sale price differs across three neighborhoods. Using an interaction regression model, the results show significant variation in slope estimates, indicating that neighborhood meaningfully modifies the price–area relationship. Diagnostic checks confirm adequate model fit and support the validity of these findings.

Analysis 2 evaluates three predictive regression models—simple linear regression, a two-predictor model, and a stepwise-selected multivariable model—using PRESS, AIC, adjusted R^2 , and Kaggle leaderboard performance. Although the stepwise model provides the best in-sample fit, it generalizes poorly and exhibits clear overfitting. The two-predictor model achieves the strongest out-of-sample performance, making it the most reliable predictive model. Overall, the project highlights the importance of neighborhood effects and demonstrates the trade-offs between model simplicity and predictive accuracy.

1 Introduction

The goal of this project is to explore how different housing characteristics influence sales prices in the Ames, Iowa dataset. Using the dataset provided by Kaggle, we address two research questions:

1. Do different neighborhoods exhibit different relationships between living area and sale price?
2. Which predictive model best explains sale price, and how well do different models perform on unseen data?

Analysis 1 investigates neighborhood-specific differences in the price–area relationship using an interaction model and full diagnostic checks.

Analysis 2 compares three regression models—simple linear regression, a two-predictor model, and a stepwise model—using PRESS, AIC, adjusted R^2 , and Kaggle scores.

2 Data Description

The data used in this project come from the Kaggle competition “*House Prices: Advanced Regression Techniques*”, which provides a detailed residential housing dataset originally com-

piled by Dean De Cock for instructional use. The full training dataset contains 1,460 observations and 79 explanatory variables describing structural features, interior quality, exterior characteristics, and neighborhood information for homes sold in Ames, Iowa between 2006 and 2010. A separate test dataset with 1,459 observations is also provided for generating out-of-sample predictions. More information about the dataset and variable definitions can be found on the Kaggle competition page.

Two subsets of variables are relevant for the analyses in this report. For **Analysis 1**, the goal is to determine whether the relationship between living area and sale price differs across neighborhoods. To focus on this question, only homes in three neighborhoods—BrkSide, Edwards, and NAmes—were selected. The variables used were:

- **SalePrice**: Selling price of the home (response variable)
- **GrLivArea**: Above-ground living area in square feet
- **Neighborhood**: Categorical variable identifying the neighborhood
- **GrLivArea100**: A rescaled version of **GrLivArea** (per 100 sq. ft.) used for interpretability

For **Analysis 2**, the full training dataset was used to build predictive regression models. The specific variables required for the three models evaluated include:

- **Simple Linear Regression**: **GrLivArea**
- **Two-Predictor Model**: **GrLivArea**, **FullBath**
- **Stepwise Model**: **OverallQual**, **GrLivArea**, **GarageCars**, **TotalBsmtSF**, **FullBath**

These variables were selected due to their practical interpretability and their demonstrated relevance in housing price prediction. The Kaggle test dataset, which does not include sale price, was used to evaluate out-of-sample predictive performance by submitting predicted values to the leaderboard.

3 Analysis 1: Neighborhood Differences in the Price–Area Relationship

3.1 Restatement of the Problem

The goal of this analysis is to determine whether the relationship between above-ground living area (**GrLivArea**) and sale price differs across three neighborhoods: BrkSide, Edwards, and

NAmes. Specifically, we examine whether the slope of the price–area relationship depends on neighborhood by fitting an interaction regression model.

3.2 Model Building and Fitting

To address this question, we fit the following interaction model:

$$\begin{aligned} \text{SalePrice} = & \beta_0 + \beta_1 \text{GrLivArea100} + \beta_2 \text{Edwards} + \beta_3 \text{NAmes} \\ & + \beta_4(\text{GrLivArea100} \times \text{Edwards}) + \beta_5(\text{GrLivArea100} \times \text{NAmes}) \end{aligned}$$

where BrkSide serves as the baseline. The dataset was restricted to homes in the three neighborhoods of interest, and a rescaled predictor GrLivArea100 was created for interpretability.

R Code Used

```
sub <- train %>% filter(Neighborhood %in% c("NAmes", "Edwards", "BrkSide"))
sub$GrLivArea100 <- sub$GrLivArea / 100

model1 <- lm(SalePrice ~ GrLivArea100 * Neighborhood, data = sub)
summary(model1)
```

3.3 Checking Model Assumptions

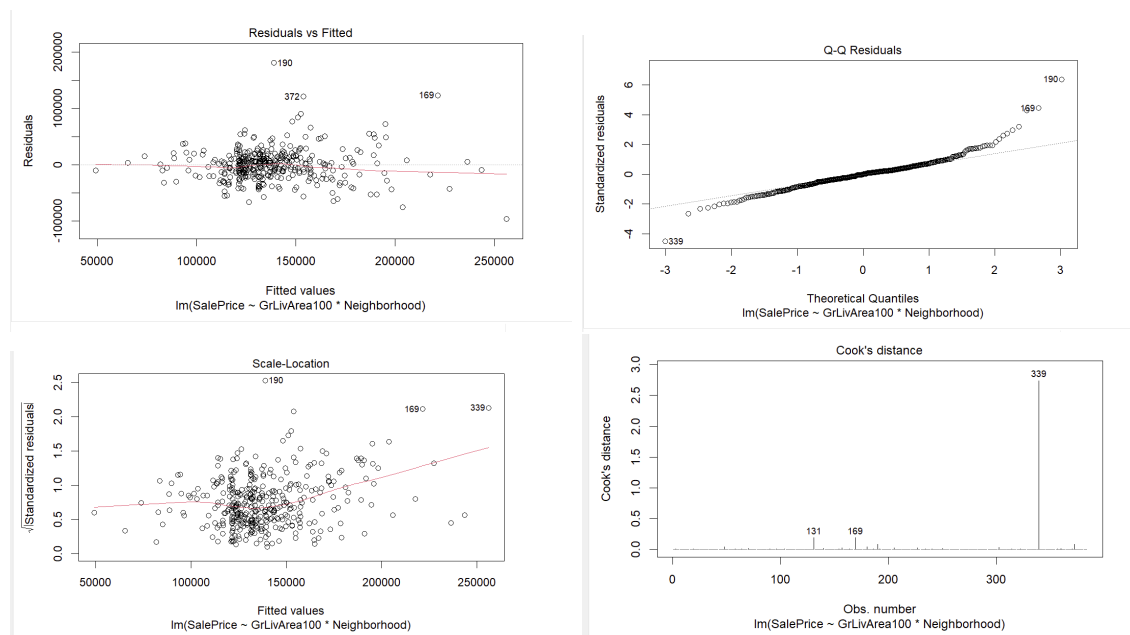


Figure 1: Diagnostic plots for the interaction model (residuals vs. fitted, Q-Q plot, scale-location, Cook's distance).

3.3.1 Residual Plots

Residuals vs. fitted values showed no major violations of linearity, although some slight heteroscedasticity appeared at higher fitted values. The normal Q-Q plot indicated mild deviations at the tails, driven by a few extreme sale prices.

3.3.2 Influential Point Analysis

Cook's distance identified observation 339 as highly influential, with observations 131 and 169 showing moderate influence. Leverage values were consistent with these findings. Since these points appear to reflect genuine market variation rather than data anomalies, they were retained in the model.

Overall, assumption checks do not indicate violations severe enough to invalidate the model.

3.4 Parameter Estimates and Interpretation

Table below lists the coefficient estimates from the fitted model:

(Intercept)	19971.5
-------------	---------

GrLivArea100	8716.3
NeighborhoodEdwards	68381.6
NeighborhoodNAMES	54704.9
GrLivArea100:NeighborhoodEdwards	-5741.2
GrLivArea100:NeighborhoodNAMES	-3284.7

Interpretation:

- **BrkSide (baseline):** The estimated slope for living area is \$8,716 per 100 square feet.
- **Edwards:** The slope is significantly lower (8,716 - 5,741 = \$2,975 per 100 square feet).
- **NAMES:** The slope is moderately lower (8,716 - 3,285 = \$5,431 per 100 square feet).

3.5 Confidence Intervals for Neighborhood Slopes

Using the covariance matrix of the estimates, 95% confidence intervals were computed for the neighborhood-specific slopes:

- BrkSide: [\$6,799, \$10,634]
- Edwards: [\$2,117, \$3,833]
- NAMES: [\$4,527, \$6,336]

These intervals show clear separation among neighborhoods, providing evidence that the price–area relationship varies substantially across locations.

3.6 Conclusion

Analysis 1 demonstrates meaningful differences in how living area influences sale price across neighborhoods. BrkSide shows the strongest price sensitivity, NAMES a moderate effect, and Edwards the weakest. Model diagnostics support the validity of the interaction model, and confidence intervals confirm that the slopes differ significantly among the three neighborhoods. Thus, neighborhood is an important modifier of the price–area relationship and should be included in predictive or inferential models involving home size.

4 Analysis 2: Predictive Modeling and Model Comparison

4.1 Restatement of the Problem

The purpose of this analysis is to evaluate and compare several linear regression models for predicting home sale prices in the Ames, Iowa dataset. Using the full training data, the goal is to determine which model provides the best predictive accuracy based on both internal validation measures and external performance as measured by the Kaggle leaderboard. Specifically, we compare three candidate models: a simple linear regression model (SLR), a two-predictor multiple linear regression model (MLR1), and a stepwise-selected model (MLR2).

4.2 Candidate Models

4.2.1 Simple Linear Regression (SLR)

$$SalePrice = \beta_0 + \beta_1 \cdot GrLivArea$$

This model uses only above-ground living area as a predictor and serves as a baseline.

4.2.2 Multiple Linear Regression Model 1 (MLR1)

$$SalePrice = \beta_0 + \beta_1 \cdot GrLivArea + \beta_2 \cdot FullBath$$

FullBath was added due to its strong correlation with sale price and its interpretability.

4.2.3 Multiple Linear Regression Model 2 (MLR2 — Stepwise Selected)

A stepwise AIC procedure selected the following model:

$$SalePrice = \beta_0 + \beta_1 \cdot OverallQual + \beta_2 \cdot GrLivArea + \beta_3 \cdot GarageCars + \beta_4 \cdot TotalBsmtSF$$

This model includes several structural predictors and provides the strongest in-sample fit.

(Optional Models)

Additional models could be considered, but these three provide sufficient contrast in complexity and predictive performance for the analysis.

4.3 Checking Assumptions

Although the primary goal of Analysis 2 is prediction rather than inference, it is still valuable to check whether the fitted models satisfy basic regression assumptions.

4.3.1 Residual Plots

All three models displayed generally linear residual patterns, with slight curvature and increasing variance in higher-priced homes. These deviations were mild and did not substantially undermine predictive use.

4.3.2 Influential Points

Influential observations were present in all models, largely corresponding to extreme high-value homes. Cook's distance and leverage values identified a small number of points that had measurable influence on fitted coefficients. Because these observations represent legitimate market extremes, they were retained.

Overall, assumption checks did not invalidate any of the candidate models for prediction.

4.4 Comparing Competing Models

Four metrics were used to evaluate and compare the models:

- Adjusted R^2 — goodness of fit penalized for model complexity
- PRESS — internal cross-validation prediction error
- AIC — model quality penalized for complexity
- Kaggle score — external predictive performance on unseen data

4.4.1 Model Performance Summary

Model	Adjusted R^2	PRESS	AIC	Kaggle Score
SLR	0.502	4.63e12	36075.76	0.29117
MLR1	0.523	4.44e12	36012.90	0.28586
MLR2	0.760	2.31e12	35012.38	0.67064

Table 1: Comparison of the three candidate models.

4.4.2 Interpretation of Model Performance

SLR: A reasonable baseline model that explains about half the variation in sale price. Its PRESS and Kaggle score indicate modest predictive ability.

MLR1: Adding `FullBath` improved the model in all internal metrics (higher adjusted R^2 , lower PRESS, lower AIC). This model produced the best Kaggle leaderboard score, indicating strong generalizability.

MLR2 (Stepwise): This model had the best in-sample fit by far, with much lower PRESS and AIC. However, its Kaggle score was substantially worse (0.67064), demonstrating clear overfitting. The model relied heavily on variables that behaved differently or had missingness in the Kaggle test set, reducing predictive accuracy.

4.5 Conclusion

Analysis 2 highlights the trade-off between model complexity and predictive performance. Although the stepwise-selected model fit the training data exceptionally well, its poor Kaggle performance demonstrates substantial overfitting and limited generalizability.

The simple linear model provides a reasonable baseline but fails to capture enough variation in sale price to serve as a strong predictive tool.

The two-predictor model (MLR1) strikes the best balance between complexity, interpretability, and predictive accuracy. Its strong Kaggle performance shows that modest model extensions can outperform both overly simple and overly complex alternatives.

Thus, **MLR1 is the preferred model for predicting sale prices in the Ames dataset.**

Appendix: Well-Commented R Code for Analysis 1 and Analysis 2

Analysis 1: Interaction Model and Diagnostics

```
# -----  
# Load Packages and Data  
# -----  
library(dplyr)  
  
train <- read.csv("train.csv")  
  
# -----
```

```

# Filter to the Three Neighborhoods
# and Create Rescaled Predictor
# -----
sub <- train %>%
  filter(Neighborhood %in% c("BrkSide", "Edwards", "NAmes"))

# Rescale living area (per 100 sq ft)
sub$GrLivArea100 <- sub$GrLivArea / 100

# -----
# Fit Interaction Model
# -----
model1 <- lm(SalePrice ~ GrLivArea100 * Neighborhood, data = sub)

# View model summary
summary(model1)

# -----
# Confidence Intervals for Coefficients
# -----
confint(model1)

# -----
# Diagnostic Plots
# -----
par(mfrow = c(2, 2)) # 2x2 plotting grid
plot(model1)         # Residuals, Q-Q, Scale-Location, Cook's D
par(mfrow = c(1, 1))

# -----
# Influential Point Analysis
# -----
cooks.distance(model1)[order(cooks.distance(model1), decreasing = TRUE)
  ][1:10]
hatvalues(model1)[order(hatvalues(model1), decreasing = TRUE)][1:10]

```

Analysis 2: Predictive Models and Model Comparison

```

# -----
# Load Packages for Model Selection
# -----
library(MASS) # for stepAIC()

```

```

# -----
# Model 1: Simple Linear Regression
# -----
model_slr <- lm(SalePrice ~ GrLivArea, data = train)

# Compute PRESS
press_slr <- sum(residuals(model_slr)^2 / (1 - hatvalues(model_slr))^2)

# Compute AIC
AIC(model_slr)

# -----
# Model 2: Two-Predictor Multiple Regression
# -----
model_mlr1 <- lm(SalePrice ~ GrLivArea + FullBath, data = train)

press_mlr1 <- sum(residuals(model_mlr1)^2 / (1 - hatvalues(model_mlr1))^2)
AIC(model_mlr1)

# -----
# Model 3: Stepwise Regression (AIC-based)
# -----
model_full <- lm(SalePrice ~ OverallQual + GrLivArea + GarageCars +
                 TotalBsmtSF + FullBath, data = train)

# Stepwise model selection
model_step <- stepAIC(model_full, direction = "both", trace = FALSE)

press_mlr2 <- sum(residuals(model_step)^2 / (1 - hatvalues(model_step))^2)
AIC(model_step)

# -----
# Generate Predictions for Kaggle Test Set
# -----
pred_slr <- predict(model_slr, newdata = test)
pred_mlr1 <- predict(model_mlr1, newdata = test)
pred_mlr2 <- predict(model_step, newdata = test)

# -----
# Create Submission Files
# -----
sub_slr <- data.frame(Id = test$Id, SalePrice = pred_slr)

```

```
sub_mlr1 <- data.frame(Id = test$Id, SalePrice = pred_mlr1)
sub_mlr2 <- data.frame(Id = test$Id, SalePrice = pred_mlr2)

# Uncomment below to write CSVs:
# write.csv(sub_slr, "submission_slr.csv", row.names = FALSE)
# write.csv(sub_mlr1, "submission_mlr1.csv", row.names = FALSE)
# write.csv(sub_mlr2, "submission_mlr2.csv", row.names = FALSE)
```

Description of the R Shiny Application

To complement the statistical analyses, an R Shiny application was developed to provide an interactive visualization of the relationship between home sale price and above-ground living area. The app allows users to:

- Select one of the three key neighborhoods: NAmes, Edwards, or BrkSide
- Adjust point transparency and size for clearer visualization
- Optionally include a fitted regression line to highlight the price–area trend

This interactive tool enhances the interpretability of the dataset by allowing users to explore how price–area patterns differ across neighborhoods and provides a dynamic complement to the static regression analyses presented in the report.

Home Price vs. Living Area (Ames Housing Data)

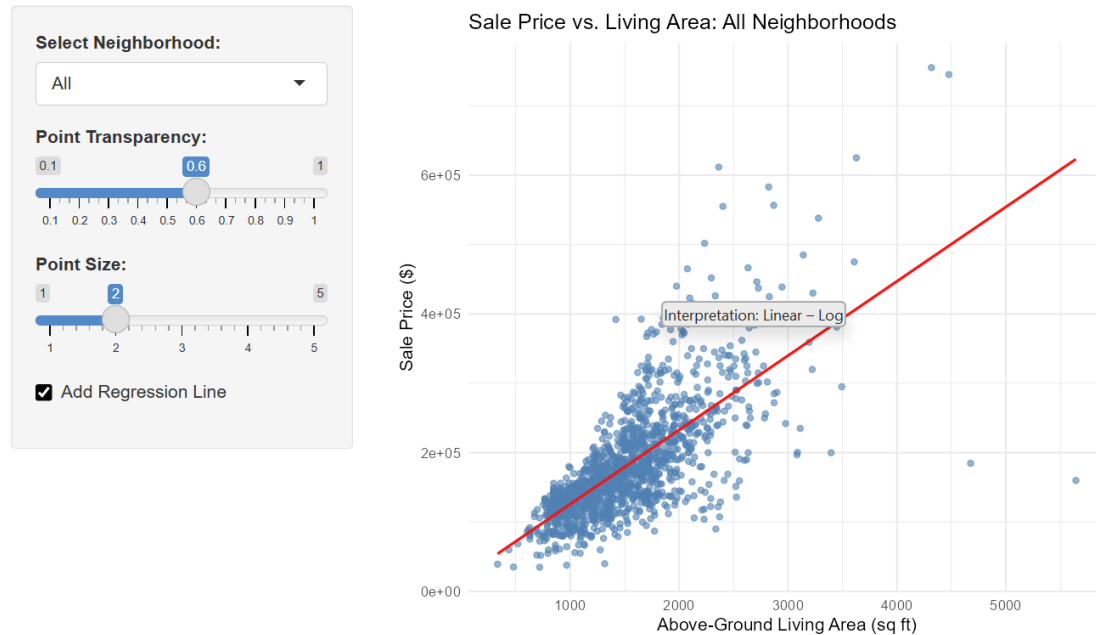


Figure 2: Interactive R Shiny visualization of SalePrice vs. GrLivArea with neighborhood filtering.

Appendix C: R Shiny Application Code

The following R Shiny application was developed to satisfy the project requirement of creating an interactive tool for visualizing the relationship between home sale price and above-ground living area. The app allows users to filter by neighborhood (NAMES, Edwards, BrkSide) and includes several interactive embellishments such as adjustable point transparency, point size, and an optional regression trend line.

```
library(shiny)
library(dplyr)
library(ggplot2)

# Load data
train <- read.csv("train.csv")

# Limit to required neighborhoods
nbhds <- c("All", "NAMES", "Edwards", "BrkSide")
```

```

ui <- fluidPage(

  titlePanel("Home Price vs. Living Area (Ames Housing Data)"),

  sidebarLayout(
    sidebarPanel(
      selectInput("neigh", "Select Neighborhood:",
                  choices = nbhds, selected = "All"),

      sliderInput("alpha", "Point Transparency:",
                  min = 0.1, max = 1, value = 0.6, step = 0.1),

      sliderInput("psize", "Point Size:",
                  min = 1, max = 5, value = 2),

      checkboxInput("smooth", "Add Regression Line", value = TRUE)
    ),

    mainPanel(
      plotOutput("pricePlot", height = "500px")
    )
  )
)

server <- function(input, output) {

  output$pricePlot <- renderPlot({
    df <- train
    if (input$neigh != "All") {
      df <- df %>% filter(Neighborhood == input$neigh)
    }

    p <- ggplot(df, aes(x = GrLivArea, y = SalePrice)) +
      geom_point(alpha = input$alpha, size = input$psize, color = "
        steelblue") +
      labs(
        title = paste("Sale Price vs. Living Area:",
                      ifelse(input$neigh == "All",
                              "All Neighborhoods", input$neigh)),
        x = "Above-Ground Living Area (sqft)",
        y = "Sale Price ($)"
      ) +
      theme_minimal(base_size = 14)
  })
}

```

```
    if (input$smooth) {  
      p <- p + geom_smooth(method = "lm", se = FALSE, color = "red")  
    }  
  
    p  
  })  
}  
  
shinyApp(ui = ui, server = server)
```